

SPM622

Project & Dissertation



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GEDION BASSON, MOE GREEN, DYLAN SHEPARD, SIHLE KAYISE
TrackWIL

Team Details

Project Title	TrackWIL
Group Members	Willem Basson, Dylan Shepard, Moegammad Green, Sihle Kayise
Module	SPM622
Lecturer / Supervisor	Amenda Raganya
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GitHub Link

https://github.com/Noobey32/TrackWIL

Table of Contents

Team Details 2
GitHub Link 2
CHAPTER 1 - INTRODUCTION 5
1.1 Background of the Study 5
1.2 Problem Statement 5
1.3 Research Aim 6
1.4 Research Objectives 6
1.5 Research Questions 6
1.6 Proposed Software Solution 6
1.7 Scope 7
1.8 Significance of the Study 9
1.9 Chapter Summary 10
CHAPTER 2 - LITERATURE REVIEW 11
2.1 Introduction 11
2.2 Existing Research 11
2.3 Existing or Similar Systems 12
2.4 Technology Review 13
2.5 Research Gap 14
2.6 Conceptual / Technical Direction 14
2.7 Chapter Summary 15
Reference list 16
CHAPTER 3 - REQUIREMENTS & METHODOLOGY 17
3.1 Introduction 17
3.2 Stakeholders and Users 17
3.3 Functional Requirements 18
3.4 Non-Functional Requirements 19
3.5 Requirements Gathering 19
3.6 Development Methodology 21
3.7 Use Cases / User Stories 21
3.8 System Architecture 22
3.9 Ethical, Privacy and Security Considerations 23
3.10 Chapter Summary 24
SOFTWARE REQUIREMENTS SPECIFICATION (SRS) 25
1. Purpose and Scope 25
2. Users and Stakeholders 25
3. Functional Requirements 26
4. Non-Functional Requirements 27
5. Use Cases / User Stories 27

6. Data Requirements 28
7. Interfaces 28
8. Constraints and Assumptions 28
9. Acceptance Criteria 29
CHAPTER 4 - SYSTEM DESIGN & DEVELOPMENT 30
4.1 Introduction 30
4.2 System Architecture 30
4.3 UML Design 32
4.4 Database Design 33
4.5 User Interface Design 34
4.6 Development Environment 34
Development Evidence 35
Revised Chapters 1 and 2 36
WEEKLY PROJECT PROGRESS REPORT 37
Week / Date 37
Planned Work 37
Completed Work 37
Challenges / Issues 39
Next Week Plan 39
Supervisor Feedback 39
Sign-off 39
PROJECT CHARTER & WORK PLAN 40
Project Overview 40
Aim and Objectives 40
Scope and Deliverables 40
Team and Roles 40
Milestones 41
Risks and Assumptions 41
Approval 41
Work plan 42

CHAPTER 1 - INTRODUCTION

1.1 Background of the Study

Work-Integrated Learning (WIL) is an important part of many occupational and technical qualifications because it gives students an opportunity to gain practical workplace experience and apply the knowledge and skills they have learned during their studies. During a WIL placement, students are required to complete workplace activities, record the work they have done, complete daily documentation and provide evidence of their activities.

Currently, the management of WIL activities can involve students completing documents daily and sending them to their workplace mentors or supervisors for verification and signing. Students then must wait for the signed documents to be returned before they can continue with the submission or verification process. These documents may be exchanged through email or other communication methods, which can make it difficult for students and academic staff to keep track of all required documents.

When documents are stored in different locations or sent between students and mentors, there is also a risk that documents may be misplaced, accidentally deleted or not returned on time. Coordinators and lecturers may also have difficulty determining which students have completed their daily requirements and which students are still waiting for documentation to be completed.

TrackWIL is proposed as a digital solution to improve this process. The system will provide students with a central platform where they can complete their daily WIL activities, submit their documentation and upload evidence. Workplace mentors will be able to access and verify the relevant submissions, while coordinators and lecturers will be able to monitor student progress.

1.2 Problem Statement

The current WIL documentation process can be time-consuming because students are required to complete daily documents and send them to their workplace mentors for verification and signing. Students then must wait for the signed documents to be returned. When this process is managed through emails, separate files or other manual methods, documents can be misplaced, accidentally deleted or returned late.

This affects students, because they need to keep track of their daily documents and ensure that they are completed and signed by their mentors. Delays in receiving signed documents can also make it difficult for students to keep their WIL records up to date.

It also affects WIL coordinators, who need to monitor whether students are completing their required activities and receiving the necessary verification. Checking documents from multiple students can become difficult and time-consuming.

Lecturers are also affected because they need to identify which students are up to date and which students are falling behind. Without a centralised system, it can be difficult to obtain an accurate and current view of each student's WIL progress.

1.3 Research Aim

The aim of this project is to design and develop TrackWIL, a centralized digital WIL management application that allows students to record their daily workplace activities, submit documentation and evidence for mentor verification, and enables coordinators and lecturers to monitor student progress.

1.4 Research Objectives

- Analyse the current WIL documentation and monitoring process to identify the challenges experienced by students, workplace mentors, coordinators and lecturers.
- Design an integrated WIL management system that allows students to submit daily WIL documents and evidence to their workplace mentors for verification.
- Develop the TrackWIL application with separate dashboards for students, mentors, coordinators and lecturers.
- Implement features that allow users to submit, verify, store and monitor WIL documents and daily activities digitally.
- Test and evaluate the TrackWIL application to determine whether it improves the efficiency of WIL documentation, verification and progress monitoring.

1.5 Research Questions

1. What challenges are experienced by students, workplace mentors, coordinators and lecturers when managing WIL documentation and activities using the current process?
2. How can an integrated digital system improve the submission and verification of students' daily WIL documents?
3. How can TrackWIL support students, mentors, coordinators and lecturers through role-specific dashboards?
4. How can digital submission, verification, storage and monitoring of WIL documents improve the management of student WIL activities?
5. To what extent can TrackWIL improve the efficiency of WIL documentation, verification and student progress monitoring?

1.6 Proposed Software Solution

The proposed software solution is TrackWIL, a web-based application designed to improve the management of Work-Integrated Learning activities and documentation.

The system will provide different dashboards based on the user's role. Students will have a dashboard where they can record their daily WIL activities, complete their required documents and upload evidence of the work they have completed. They will then be able to submit their documents through the system to their workplace mentor for verification.

Workplace mentors will have access to the relevant student submissions. They will be able to review the students' daily documentation, verify the information and approve the required documents. The student will then be able to see whether the document has been approved, rejected or is still waiting for the mentor's response.

WIL coordinators will have a dashboard where they can monitor students and see whether their daily documentation has been submitted and verified. They will also be able to identify outstanding documents and students who may require attention.

Lecturers will have a monitoring dashboard that provides an overview of student progress. They will be able to see which students are up to date, which students have outstanding work, and which students may be falling behind.

1.7 Scope

The TrackWIL project will include the design and development of a web-based WIL management system with the following functionality:

Student functionality

- Student registration and login.
- Student dashboard.
- Recording of daily WIL activities.
- Completion of daily WIL documentation.
- Uploading of work and supporting evidence.
- Submission of documents to workplace mentors.
- Viewing the status of submitted documents.
- Viewing mentor feedback or verification.
- Monitoring personal WIL progress.

Mentor functionality

- Mentor login.

- Viewing assigned students.
- Viewing student daily submissions.
- Reviewing uploaded documents and evidence.
- Approving or rejecting submissions.
- Verifying required documents.
- Providing feedback to students.

Coordinator functionality

- Coordinator login.
- Viewing registered WIL students.
- Monitoring daily submissions.
- Checking whether documents have been verified.
- Identifying missing or outstanding submissions.
- Monitoring student WIL progress.

Lecturer functionality

- Lecturer login.
- Viewing student progress.
- Identifying students who are up to date or behind.
- Viewing outstanding WIL activities and documentation.
- Monitoring student performance and progress.

System functionality

- User authentication.
- Role-based access.
- Document upload and storage.
- Submission tracking.

- Verification/status tracking.
- Progress monitoring.
- Basic reporting.

1.8 Significance of the Study

The TrackWIL project is expected to provide benefits to several groups involved in the WIL process.

Students

TrackWIL will make it easier for students to manage their daily WIL documentation. Instead of relying mainly on sending documents through email and waiting for them to be returned, students will have one central place to submit their documents and check their verification status.

Students will also be able to see which documents have been completed and which are still outstanding. This can help students stay organized and keep track of their WIL requirements.

Workplace Mentors

Mentors will have easier access to the documents submitted by students assigned to them. They will be able to review, verify and sign documents through the system rather than relying on separate communication channels.

This can make the verification process more organized and provide a record of completed mentor approvals.

WIL Coordinators

Coordinators will have a centralized view of student submissions and verification statuses. Instead of manually checking different documents or asking students for updates, coordinators can use the dashboard to identify missing or outstanding work.

This can reduce the administrative workload and make monitoring more efficient.

Lecturers

Lecturers will be able to monitor student progress from one dashboard. They can identify students who are up to date and students who are falling behind with their WIL requirements.

This can allow lecturers to identify problems earlier and provide support where necessary.

Educational Institutions

TrackWIL can help educational institutions improve the organization and management of WIL documentation. Centralizing information can reduce dependence on emails, spreadsheets and separate documents and provide a clearer record of student WIL progress.

Software Engineering Field

The project demonstrates how software engineering can be used to solve a real-world problem in education and workplace training. It will apply concepts such as requirements analysis, database design, user authentication, role-based access, document management, system development and software testing.

1.9 Chapter Summary

This chapter introduced the TrackWIL project and discussed the challenges associated with the current management of Work-Integrated Learning documentation and activities. Students currently need to complete their daily WIL documents, submit them to workplace mentors for verification or signing and wait for the completed documents to be returned. This process can make it difficult to track submissions and identify outstanding documentation.

The chapter presents the research aim, objectives and research questions that will guide the development of TrackWIL. The proposed solution is a centralized web-based application that will allow students to submit daily WIL documents and evidence, while workplace mentors will be able to review and verify these submissions. Coordinators and lecturers will be able to monitor student submissions and overall progress through their respective dashboards.

The scope of the project was also defined by identifying the features that will be included and excluded from the system. Finally, the significance of the study explained how TrackWIL can benefit students, workplace mentors, WIL coordinators, lecturers and educational institutions.

CHAPTER 2 - LITERATURE REVIEW

2.1 Introduction

This chapter reviews research, existing software and relevant technologies related to Work-Integrated Learning (WIL) documentation and progress monitoring. The review considers how digital systems support students, workplace mentors, coordinators and lecturers, and identifies features that can inform the design of TrackWIL.

The chapter compares three existing systems: InPlace, Sonia Online and PebblePad/Mahara-style ePortfolio tools, against the project's needs. It also considers cloud delivery, database-backed web applications, role-based access and secure document upload. The findings are used to define a practical research gap and guide TrackWIL's proposed technical direction

2.2 Existing Research

Work-Integrated Learning connects academic learning with practical workplace experience. Technology can support WIL in different ways: it may deliver learning activities, or it may provide administrative support for placements, communication, evidence collection and assessment. Schuster and Glavas (2017) explain that electronic WIL covers a range of technology uses rather than one single model. For TrackWIL, the administrative role is especially relevant because the project focuses on daily activity records, evidence submission, mentor verification and progress monitoring.

Effective placement supervision depends on communication and coordination between students and supervisors. Hains-Wesson et al. (2026), in a systematic literature review of placement supervision, examine student and supervisor perspectives and identify patterns and gaps in the literature. This supports the need to consider both sides of the submission and feedback process rather than designing a system only for students or academic staff.

Industry systems show that placement management commonly involves centralized records, user portals, communication, compliance and reporting. InPlace describes tools for central placement visibility, student portals, time sheets, communications, compliance and analytics (InPlace Software, n.d.). Sonia Online describes itself as a placement management system containing placement information and paperwork for students and host organizations (Sonia Online, n.d.). These examples show that digital coordination is an established approach, although their exact configuration and available functions depend on the institution using them.

Portfolio platforms also provide relevant ideas for collecting and reviewing evidence. Mahara allows learners to upload evidence, reflect on experiences, share portfolios with selected people and receive feedback (Mahara, n.d.). PebblePad's ATLAS assessment area supports assessors in accessing and reviewing learners' work (PebblePad, 2026). These features are relevant to TrackWIL's evidence and

mentor-review workflow, but a portfolio or assessment platform is not necessarily designed around a specific institution's daily WIL forms and status-tracking requirements.

Overall, the literature and industry examples support a centralized digital workflow, clear communication, evidence management and timely feedback. However, TrackWIL must still be evaluated against its specific project requirements, including daily submissions, mentor approval or rejection, feedback, role-specific dashboards and a visible list of outstanding work. Product websites describe vendor-provided capabilities; they should not be treated as independent proof that a system will improve outcomes in every setting.

2.3 Existing or Similar Systems

Three relevant systems were reviewed: InPlace, Sonia Online and Mahara/PebblePad. The comparison below is based on publicly available product information, not on hands-on testing. Features may vary by institution, subscription or local configuration.

System 1: InPlace: InPlace is designed for higher-education placement management. Its stated features include centralized placement management, student portals, timesheets and schedules, communications, compliance management, integrations and reporting (InPlace Software, n.d.). Its strength is broad administration and visibility across placements. A limitation for this comparison is that its public description emphasizes placement allocation, compliance and programme-level management; it does not establish that every TrackWIL-specific daily logbook or mentor-signing workflow is available in a particular deployment. Lesson for TrackWIL: centralize records and provide role-appropriate views, but prioritize the daily documentation workflow required by the project.

System 2: Sonia Online: Sonia Online is presented as a student placement management system for students and host organizations, with placement information and paperwork available through the system (Sonia Online, n.d.). Its strength is that it directly targets placement coordination and provides access for relevant placement participants. Its limitations cannot be fully assessed from the public landing page alone, and access or features may differ between institutions. Lesson for TrackWIL: make student and mentor access straightforward and keep placement paperwork in one organized location.

System 3: Mahara / PebblePad: Mahara supports evidence-based portfolios, reflection, sharing and feedback (Mahara, n.d.). PebblePad's ATLAS area supports access to learner submissions and assessment (PebblePad, 2026). Their strengths are evidence collection, sharing and review. A limitation relative to TrackWIL is that portfolio and assessment workflows may not, by themselves, provide the project's complete set of daily WIL status controls, outstanding-document monitoring and separate coordinator and lecturer dashboards. Lesson for TrackWIL: connect each uploaded evidence item to a dated activity, submission status, mentor feedback and progress overview.

Comparative summary: InPlace is strongest as a broad placement-management platform; Sonia Online is a directly relevant placement-information and paperwork system; Mahara/PebblePad provide useful portfolio and assessment patterns. TrackWIL draws on these ideas but will focus its scope on

daily WIL activity records, supporting evidence, mentor verification, feedback and monitoring by coordinators and lecturers. This is a project-specific design rationale, not a claim that the existing products lack all such features.

2.4 Technology Review

Web-based application: A browser-based application is suitable because students, mentors, coordinators and lecturers may access the system from different locations and devices. A shared application can present role-specific dashboards while keeping records in a central service. Its disadvantages include dependence on network availability, the need to support different screen sizes and the need for secure deployment and maintenance.

Database: A relational database is appropriate for structured relationships between users, roles, students, assigned mentors, daily activity entries, uploaded evidence, review comments and submission statuses. Using linked tables and primary/foreign keys can help preserve consistency and make it possible to report on missing submissions and verification progress. The final database technology should be selected to fit the group's development environment and deployment requirements; Chapter 1 does not specify a final database product.

Authentication and role-based access: TrackWIL handles student records and workplace evidence, so users should authenticate before accessing protected information. Authorization should restrict users to the records and actions needed for their roles, for example, a mentor should access assigned students' submissions, while coordinators and lecturers should have monitoring views appropriate to their responsibilities. OWASP describes access control as mediating access to resources based on identity and policy (OWASP Foundation, n.d.). Role checks should be enforced by the server, not only by hiding interface elements.

Document upload and storage: Evidence upload is a core feature, but uploaded files create security and storage risks. OWASP recommends allowing only necessary file types, validating file types rather than trusting a supplied content-type value, limiting file size, generating safe stored filenames, and ensuring only authorized users can access uploads (OWASP Foundation, n.d.). TrackWIL should also use protected storage, controlled download links, appropriate backups and clear rules for retention and deletion.

Cloud hosting: Cloud hosting can make a web application accessible over the internet without the project team maintaining all underlying physical infrastructure. NIST describes cloud computing as on-demand network access to a shared pool of configurable computing resources (Mell and Grance, 2011). Hosting decisions should still consider cost, availability, backup and recovery, privacy, access controls and the location where student information is stored. Cloud hosting is a possible direction rather than a confirmed choice in Chapter 1.

Usability and testing: The application should make submission status easy to understand and minimise unnecessary steps for students and mentors. Testing should cover successful and unsuccessful logins, role permissions, document upload validation, submission and review status

changes, feedback, and progress reports. Evaluations should check whether users can complete these tasks reliably; claims of improved efficiency should be based on testing or user feedback rather than assumed in advance.

2.5 Research Gap

Existing placement-management systems and portfolio platforms already provide useful functions such as centralized placement information, paperwork, evidence sharing, assessment, communication and reporting. Therefore, the research gap is not that no digital WIL or placement tools exist.

The project-specific gap addressed by TrackWIL is the need for a focused, integrated workflow aligned with the requirements described in Chapter 1: students record daily WIL activities and upload evidence; mentors review submissions, provide feedback and approve or reject them; and coordinators and lecturers monitor verification status and identify outstanding work through role-specific dashboards.

Public product descriptions do not establish whether one existing platform, in the configuration available to the intended users, matches all these requirements in a simple workflow. The project will therefore design and test a solution against these requirements rather than claim that existing products cannot support them. The proposed contribution is a prototype tailored to the stated use case, with a shared record of submissions and their status.

The project evaluation should investigate whether users can submit and verify documentation more clearly and whether staff can identify outstanding work more easily. This gap and evaluation focus align with the aim and objectives in Chapter 1.

2.6 Conceptual / Technical Direction

TrackWIL will follow a centralized, role-based workflow. A student creates a dated WIL activity, completes the required documentation and attaches supporting evidence. The student submits the entry to the assigned workplace mentor. The mentor reviews it and either approves it or rejects it with feedback. The system records the current status so the student can see the outcome, while coordinators and lecturers use dashboards to identify submitted, pending, rejected, approved and outstanding work.

The main conceptual entities are User, Role, Student, Mentor Assignment, WIL Activity, Document/Evidence, Submission Review and Feedback. Relationships between these entities should be designed so each activity belongs to a student, relevant submissions can be reviewed by an authorized mentor, and status changes can be reported consistently. The design should avoid storing multiple conflicting copies of the same submission.

Technical priorities are usability, data integrity, access control, safe file handling, maintainability and testability. The group should choose the implementation framework, database and hosting service based on its actual development environment and skills; the source material currently available does

not confirm a specific stack. Regardless of the stack, the system should validate input, enforce permissions on the server, restrict file uploads and test each role's workflow.

2.7 Chapter Summary

This chapter reviewed research on electronic WIL and placement supervision, and compared InPlace, Sonia Online and portfolio/assessment tools such as Mahara and PebblePad. The review found useful patterns in centralized placement information, evidence collection, submission review, feedback, compliance and reporting. It also highlighted technical considerations involving relational data, authentication, role-based access, safe file uploads, hosting and usability.

The research gap is a project-specific integrated workflow that matches TrackWIL's stated needs for daily activity logging, evidence submission, mentor verification and role-specific progress monitoring. These findings inform the functional requirements and technical design in the next project stages. TrackWIL's effectiveness must be established through testing and evaluation rather than assumed.

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CHAPTER 3 - REQUIREMENTS & METHODOLOGY

3.1 Introduction

This chapter explains how the requirements for TrackWIL will be identified and how the system will be developed. TrackWIL is a proposed web-based system designed to improve the management of WorkIntegrated Learning (WIL) documents and student progress.

Currently, students may need to complete and sign daily WIL documents, send them to workplace mentors for verification and wait for the signed documents to be returned. This manual process can lead to delays, lost documents and difficulties tracking student progress.

TrackWIL aims to provide a central platform where students can submit daily work records, mentors can review and verify submissions, and coordinators and lecturers can monitor student progress. This chapter describes the stakeholders, system requirements, requirements gathering process, development methodology, user stories, system architecture and ethical considerations that will guide the development of the system.

3.2 Stakeholders and Users

Stakeholders are people or organisations affected by TrackWIL, while users are the people who interact directly with the system.

Primary stakeholders and users

Students: Students will use TrackWIL to create and submit daily WIL records, upload supporting documents and check whether their submissions have been approved or require corrections. Their main need is a simple and reliable way to manage their WIL documentation without relying entirely on paper documents and email.

Workplace Mentors: Mentors will review records submitted by students assigned to them. They will be able to verify submissions, approve correct records and request corrections when necessary. Their main need is an organised way to review and confirm student work.

Coordinators: Coordinators will monitor student submissions and identify outstanding records. They need an overview of student documentation so that they can follow up on missing submissions and help ensure that WIL requirements are being completed.

Lecturers: Lecturers will monitor student progress and identify students who may be falling behind. They need access to relevant progress information to provide academic support and follow up when necessary.

Secondary stakeholders

Training Institution: The institution requires students to complete their WIL activities and maintain the necessary documentation. TrackWIL can support the institution by making records easier to manage and monitor.

Development Team: The development team is responsible for analysing requirements, designing, developing and testing the system. The team needs clear requirements, suitable technologies and effective collaboration to deliver the project successfully.

Project Supervisor: The project supervisor oversees the academic project and checks whether the system meets the project objectives and assessment requirements.

3.3 Functional Requirements

ID	Description	Priority
FR01	The system must allow authorized users to log in using valid credentials.	High
FR02	The system must provide different access permissions for students, mentors, coordinators and lecturers.	High
FR03	Students must be able to create daily WIL records describing the work completed.	High
FR04	Students must be able to submit completed daily records for review.	High
FR05	Students must be able to upload supporting documents or evidence related to their daily records.	High
FR06	Workplace mentors must be able to view records submitted by their assigned students.	High
FR07	Workplace mentors must be able to approve records or request corrections with feedback.	High
FR08	The system must display the status of each record as pending, approved or requiring corrections.	High
FR09	Students must be able to view their previous submissions and their status.	Medium
FR10	Coordinators must be able to identify students with outstanding or missing records.	High

FR11	Lecturers must be able to monitor student progress and identify students who are falling behind.	High
FR12	The system must record the dates and times of submissions and verifications.	High
FR13	Authorized users must be able to view relevant student activity and progress information.	Medium
FR14	Users must be able to log out of their accounts securely.	High

3.4 Non-Functional Requirements

- **Usability:** TrackWIL should have a simple, understandable interface that allows users to navigate their dashboards, submit records and check document statuses easily.
- **Performance:** The system should respond to normal user actions within an acceptable time. Dashboard information and document submission functions should load without unnecessary delays. The team should define and test specific response-time targets.
- **Security:** The system must protect user accounts and information through appropriate authentication, role-based access control and secure handling of uploaded documents.
- **Reliability:** Submitted records and supporting documents should be stored reliably. Backup and recovery measures should be considered to reduce the risk of losing important information.
- **Maintainability:** The application should use an organised project structure and clear documentation so that developers can correct errors, update features and maintain the system.
- **Compatibility:** TrackWIL should work on commonly used modern web browsers and adapt to different screen sizes, including computers, tablets and mobile phones.
- **Data Integrity:** The system should maintain accurate and consistent records. Submissions must be linked to the correct students and placements, and only authorised users should be able to make permitted changes.
- **Privacy:** Personal information and WIL documents must only be accessible to authorised users who need the information to perform their responsibilities.

3.5 Requirements Gathering

Requirements gathering is the process of identifying and understanding what a system must do and what users need from it. For the TrackWIL project, requirements will be identified by analysing the project scenario, reviewing the assignment brief, identifying the system users, and discussing the required features with the project team.

Scenario Analysis

The TrackWIL scenario will be analysed to understand the problems with the current manual WIL documentation process. Students currently complete daily work documents and send them to workplace mentors for signing and verification. This process can cause delays, and documents may be lost or accidentally deleted. TrackWIL aims to reduce these problems by allowing students to submit their daily work records digitally and making it easier for authorised users to monitor and verify submissions.

Assignment Brief and Document Analysis

The project brief and assignment requirements will be reviewed to identify the main features and expectations of TrackWIL. These requirements will guide the design and development of the system, including user login, daily work record submissions, document uploads, mentor verification, and progress monitoring.

Stakeholder Analysis

The needs of the main TrackWIL users will be identified to ensure that the system supports their responsibilities.

- **Students:** Need to record daily work activities, upload supporting documents, and check the status of their submissions.
- **Workplace mentors:** Need to view students' submissions, verify completed work, and provide feedback when corrections are required.
- **Coordinators:** Need to monitor submissions and identify outstanding or missing records.
- **Lecturers:** Need to check student progress and identify students who are falling behind with their WIL requirements.

Project Team Discussions

The project team will discuss the identified requirements and agree on the features that should be included in TrackWIL. These discussions will help the team organise the requirements according to their importance and determine which features should be developed first.

Requirements Review

The identified requirements will be reviewed to ensure that they are clear, realistic, and relevant to the TrackWIL problem. The requirements will then be documented in the Software Requirements Specification (SRS) and used to guide the system design, development, and testing.

The requirements gathering process will help ensure that TrackWIL addresses the problems associated with manual WIL documentation and meets the needs of its intended users.

3.6 Development Methodology

The proposed methodology for TrackWIL is Agile. Agile is an iterative approach that allows software to be developed in smaller stages, tested regularly and improved as the project progresses.

- **Planning:** The team identifies the problem, defines the scope and agrees on the project objectives.
- **Requirements Analysis:** The team identifies the users and documents the functional and nonfunctional requirements.
- **System Design:** The team prepares the system architecture, UML diagrams, database design and user interface wireframes.
- **Development:** The team builds the system in smaller parts, starting with essential features such as login, student dashboards and daily record submission.
- **Testing:** Each developed feature is tested to identify errors and confirm that it meets its requirements.
- **Review and Improvement:** The team reviews completed work, discusses problems and improves the system based on testing results and available feedback.
- **Implementation and Maintenance:** The application is prepared for use, and problems or improvements are addressed where required by the project scope.

Justification for Agile

Agile is suitable for TrackWIL because the system contains several connected features that can be developed and tested separately. For example, the team can develop student record submission before completing the mentor verification and progress monitoring functions.

Agile allows problems to be identified early and makes it easier to improve features as the team learns more about the system. It also supports teamwork because members can work on different tasks and regularly review their progress.

3.7 Use Cases / User Stories

1. **US-01: Student Login**

As a student, I want to log in securely so that I can access my WIL records and dashboard.

2. **US-02: Daily Record Submission**

As a student, I want to create and submit a daily WIL record so that my workplace activities are documented and stored in one place.

3. **US-03: Upload Supporting Documents**

As a student, I want to upload supporting evidence so that my workplace activities can be reviewed without relying entirely on paper documents or email attachments.

4. **US-05: Mentor Verification**

As a workplace mentor, I want to review student submissions and approve them or request corrections so that the records are properly checked.

5. **US-06: Monitor Outstanding Records**

As a coordinator, I want to identify students with missing or outstanding records so that I can follow up and help ensure that the required documentation is completed.

6. **US-07: Monitor Student Progress**

As a lecturer, I want to view relevant student progress information so that I can identify students who are falling behind and provide support.

3.8 System Architecture

TrackWIL is proposed as a web-based application that connects users to a central system for managing WIL records and documents. The system architecture will be organised into three main layers: the presentation layer, application layer and data layer.

Presentation Layer

The presentation layer is the part of the system that users interact with through a web browser. It will provide the login page and dashboards for students, workplace mentors, coordinators and lecturers.

Students will use this layer to create daily records, upload evidence and check submission statuses. Mentors will use it to review and verify submissions, while coordinators and lecturers will use their dashboards to monitor student progress.

Application Layer

The application layer handles the system's main operations and business rules. These include authentication, checking user permissions, processing daily record submissions, managing document verification and updating submission statuses.

This layer must ensure that users can only perform actions permitted by their roles. For example, a student should not be able to approve their own submission as a mentor.

Data Layer

The data layer stores and manages the information required by TrackWIL. This may include user accounts, WIL placements, daily records, document references and verification history.

A relational database such as PostgreSQL could be used to maintain relationships between these records. Uploaded documents would be stored using an appropriate file storage service, with access restricted to authorised users.

Communication Between the Layers

When a student submits a daily record, the presentation layer sends the information to the application layer. The application layer checks the request, verifies the user's permissions and saves the record through the data layer. The system then returns a response so that the student can see whether the submission was successful.

When a mentor reviews the record, the same general process is used to retrieve the relevant information and save the verification decision.

This architecture separates the user interface, application operations and data storage, making the system easier to organise, test and maintain. The final architecture diagram will show these layers and the connections between them.

The exact frameworks, database and storage services will be confirmed according to the technology stack selected by the development team.

3.9 Ethical, Privacy and Security Considerations

User Authentication and Access Control

TrackWIL must use appropriate authentication and role-based permissions to protect user accounts. Students should access their own records, mentors should access records assigned to them, and coordinators and lecturers should only access information required for their responsibilities.

Data Accuracy and Accountability

The system should maintain accurate submission and verification information. Important actions, such as submitting a record, approving it or requesting corrections, should be recorded where appropriate. This will help users understand the history of a document and reduce confusion about whether it has been reviewed. Verification must also be performed by the authorised mentor rather than being treated as automatically approved.

Consent and Responsible Data Use

Users should receive appropriate information about what personal information the system collects and why it is needed. Where consent is the appropriate lawful basis for processing, it must be obtained in accordance with applicable requirements. Personal information should only be used for authorised purposes.

Privacy and Protection of Personal Information

The system should collect only the information necessary for WIL administration and monitoring. Personal information must be handled responsibly and in accordance with applicable institutional policies and South African data protection requirements, including the Protection of Personal Information Act (POPIA), where applicable.

3.10 Chapter Summary

This chapter presented the requirements and proposed development methodology for TrackWIL. It identified the main stakeholders and users, including students, workplace mentors, coordinators and lecturers, and explained their needs.

The functional requirements described the features needed for login, daily record submission, document uploads, mentor verification and progress monitoring. The non-functional requirements addressed usability, performance, security, reliability, compatibility, maintainability, data integrity and privacy.

The chapter also explained how requirements can be gathered and reviewed, justified the proposed use of Agile over Waterfall, and documented user stories and the main daily record verification use case. Finally, it described the proposed three-tier system architecture and discussed ethical, privacy and security considerations.

These requirements and design decisions will guide the detailed Software Requirements Specification, the system design in Chapter 4, and the implementation and testing of TrackWIL.

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

1. Purpose and Scope

TrackWIL is a web-based system proposed to improve the management of Work-Integrated Learning (WIL) documents and student progress. The system replaces the manual, paper-reliant process of completing and signing daily WIL documents, which often leads to lost records and processing delays. The scope of the platform includes centralizing the workflow so that students can digitally submit daily work records and evidence, workplace mentors can efficiently review and verify submissions, and academic coordinators and lecturers can seamlessly monitor overall student progress.

2. Users and Stakeholders

- **Students (Primary):** Need a simple, reliable platform to create daily WIL records, upload supporting documents, and check the verification status of their submissions without relying on email or paper.
- **Workplace Mentors (Primary):** Require an organized interface to view records submitted by assigned students, verify completed work, approve records, and request necessary corrections with feedback.
- **Coordinators (Primary):** Need an overview of student documentation to monitor submissions, identify missing records, and ensure WIL requirements are fulfilled.
- **Lecturers (Primary):** Need to track academic progression to identify students who are falling behind and provide necessary support.
- **Secondary Stakeholders:** Include the Training Institution (needs efficient record management), the Development Team (needs clear requirements and tools to build the system), and the Project Supervisor (oversees the project's academic alignment).

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3. Functional Requirements

- **FR-01:** The system must allow authorized users to log in using valid credentials. (Priority: High) | *Acceptance Criterion: User is granted access to the appropriate role-based dashboard upon entering correct credentials.*
- **FR-02:** The system must provide different access permissions for students, mentors, coordinators, and lecturers. (Priority: High) | *Acceptance Criterion: Users can only access features and data permitted by their designated roles.*
- **FR-03:** Students must be able to create daily WIL records describing the work completed. (Priority: High) | *Acceptance Criterion: The system provides an accessible form for students to input daily tasks.*
- **FR-04:** Students must be able to submit completed daily records for review. (Priority: High) | *Acceptance Criterion: Record successfully transitions to a "pending" state for mentor review upon submission.*
- **FR-05:** Students must be able to upload supporting documents or evidence related to their daily records. (Priority: High) | *Acceptance Criterion: The system accepts and securely attaches file uploads to the respective daily record.*
- **FR-06:** Workplace mentors must be able to view records submitted by their assigned students. (Priority: High) | *Acceptance Criterion: Mentors can access a populated list of pending records from their specific mentees.*
- **FR-07:** Workplace mentors must be able to approve records or request corrections with feedback. (Priority: High) | *Acceptance Criterion: The system allows mentors to attach feedback notes and update the record state accordingly.*
- **FR-08:** The system must display the status of each record as pending, approved, or requiring corrections. (Priority: High) | *Acceptance Criterion: Clear status indicators are visible on both the student and mentor dashboards.*
- **FR-09:** Students must be able to view their previous submissions and their status. (Priority: Medium) | *Acceptance Criterion: A history log is available for students to track past entries.*
- **FR-10:** Coordinators must be able to identify students with outstanding or missing records. (Priority: High) | *Acceptance Criterion: Coordinators have access to a reporting view highlighting unsubmitted mandatory records.*
- **FR-11:** Lecturers must be able to monitor student progress and identify students who are falling behind. (Priority: High) | *Acceptance Criterion: Lecturers can view aggregated metrics on student progression.*
- **FR-12:** The system must record the dates and times of submissions and verifications. (Priority: High) | *Acceptance Criterion: All record interactions generate accurate, automated timestamps.*
- **FR-13:** Authorized users must be able to view relevant student activity and progress information. (Priority: Medium) | *Acceptance Criterion: Dashboard information dynamically updates based on the viewing user's authorization level.*

- **FR-14:** Users must be able to log out of their accounts securely. (Priority: High) | *Acceptance Criterion: Active sessions are completely terminated upon logging out.*

4. Non-Functional Requirements

- **NFR-01 (Usability):** TrackWIL should feature a simple, understandable interface for seamless navigation, record submission, and status checking.
- **NFR-02 (Performance):** The system must load dashboard information and process document submissions within defined, acceptable response-time targets without unnecessary delays.
- **NFR-03 (Security):** The system must enforce role-based access control, secure document handling, and proper authentication mechanisms.
- **NFR-04 (Reliability):** TrackWIL must guarantee the reliable storage of submitted records and documents, utilizing backup and recovery measures to mitigate data loss risks.
- **NFR-05 (Maintainability):** The application will utilize an organized project structure and clear documentation to support developers in correcting errors, maintaining features, and updating the system over time.
- **NFR-06 (Compatibility):** The interface must adaptively support different screen sizes, including mobile phones, tablets, and computers, across commonly used modern web browsers.
- **NFR-07 (Data Integrity):** The system must maintain accurate data relationships, strictly linking submissions to correct placements and students, and restricting changes exclusively to authorized users.
- **NFR-08 (Privacy):** Access to WIL documents and personal information must be strictly limited to users who require the data to fulfill their specific responsibilities.

5. Use Cases / User Stories

- **US-01: Student Login:** As a student, I want to log in securely so that I can access my WIL records and dashboard.
- **US-02: Daily Record Submission:** As a student, I want to create and submit a daily WIL record so that my workplace activities are documented and stored in one place.
- **US-03: Upload Supporting Documents:** As a student, I want to upload supporting evidence so that my workplace activities can be reviewed without relying entirely on paper documents or email attachments.
- **US-05: Mentor Verification:** As a workplace mentor, I want to review student submissions and approve them or request corrections so that the records are properly checked.
- **US-06: Monitor Outstanding Records:** As a coordinator, I want to identify students with missing or outstanding records so that I can follow up and help ensure that the required documentation is completed.
- **US-07: Monitor Student Progress:** As a lecturer, I want to view relevant student progress information so that I can identify students who are falling behind and provide support.

6. Data Requirements

The system requires the management of fundamental data entities, including user accounts, WIL placements, daily activity records, document references, and chronological verification histories. Validation rules must ensure that each submission is properly linked to its corresponding student and workplace placement. Important system actions, such as submitting, approving, or requesting corrections on a record, must be documented to maintain an accurate document history. Data must be reliably retained; a relational database architecture (e.g., PostgreSQL) will be used to maintain these complex relationships, while uploaded evidence files will be held in a secure, access-restricted file storage service.

7. Interfaces

- **User Interface (UI) / Presentation Layer:** A web-based front-end that provides specific, role-based dashboards for students, mentors, coordinators, and lecturers. It will handle all direct interactions, such as record creation, file uploading, and status monitoring.
- **Application Interface / Business Logic Layer:** Sits between the UI and database to process logic, authenticate sessions, check role permissions, manage document verification rules, and prevent unauthorized actions (e.g., ensuring a student cannot approve their own submission as a mentor).
- **Database Interface / Data Layer:** Responsible for retrieving and saving records securely. When a request is validated by the application layer, this interface communicates with the relational database to commit the data and returns a success/failure response back to the user.

8. Constraints and Assumptions

- **Legal & Privacy Constraints:** The system must strictly adhere to South African data protection regulations, specifically the Protection of Personal Information Act (POPIA), ensuring that data is only collected for authorized WIL administration purposes and that user consent is handled appropriately.
- **Technical Constraints:** The development will follow an Agile System Development Life Cycle (SDLC) approach to iteratively test and implement features like login, submissions, and dashboard routing.
- **Operational Assumptions:** It is assumed that all stakeholders will have access to modern web browsers and an internet connection to utilize the cloud-based functionalities effectively. Mentors are assumed to manually verify the accuracy of work before approving, rather than relying on automated approvals.

9. Acceptance Criteria

The solution will be accepted when:

1. Students can successfully log in, fill out daily task records, upload corresponding evidence files, and transmit them for review.
2. Workplace mentors can successfully view assigned submissions, toggle statuses to approved, or return records with necessary correction feedback.
3. Coordinators and lecturers can successfully pull aggregated data to identify missing documentation and track academic pacing.
4. Role-based security is proven to restrict permissions effectively across the application layer, ensuring data privacy standards and POPIA compliance are met.
5. Iterative Agile testing confirms that all primary functional features load and operate without debilitating performance delays or data integrity loss.

CHAPTER 4 - SYSTEM DESIGN & DEVELOPMENT

4.1 Introduction

This chapter shows how we turned Chapter 3 requirements into a working design and what we built in the development sprint. TrackWIL replaces the PDF and email logbook with a small web-app. Students log workplace hours, the employer checks and entry through a link and does not need an account to do so. The WIL coordinator can see who is on track, who is waiting on a mentor, and who is overdue.

For week 3 we drew the architecture, UML, and database design. We built the app at the same time. The code in `05_Source_Control/trackwil` covers the three core features from the proposal.

4.2 System Architecture

TrackWIL is a Next.js application. The browser talks to a single Node.js process. That process serves the React pages, runs the business logic, and reads and writes to the JSON database. Uploaded evidence sits on disk.

Main components:

Component	Responsibility
Student area	Dashboard, new entries, evidence, sharing
Coordinator area	Table of students with status, verifying hours
Public verify	Mentor entry approval
Auth	Login, logout, signup
Verify action	Checks token, updates entry status

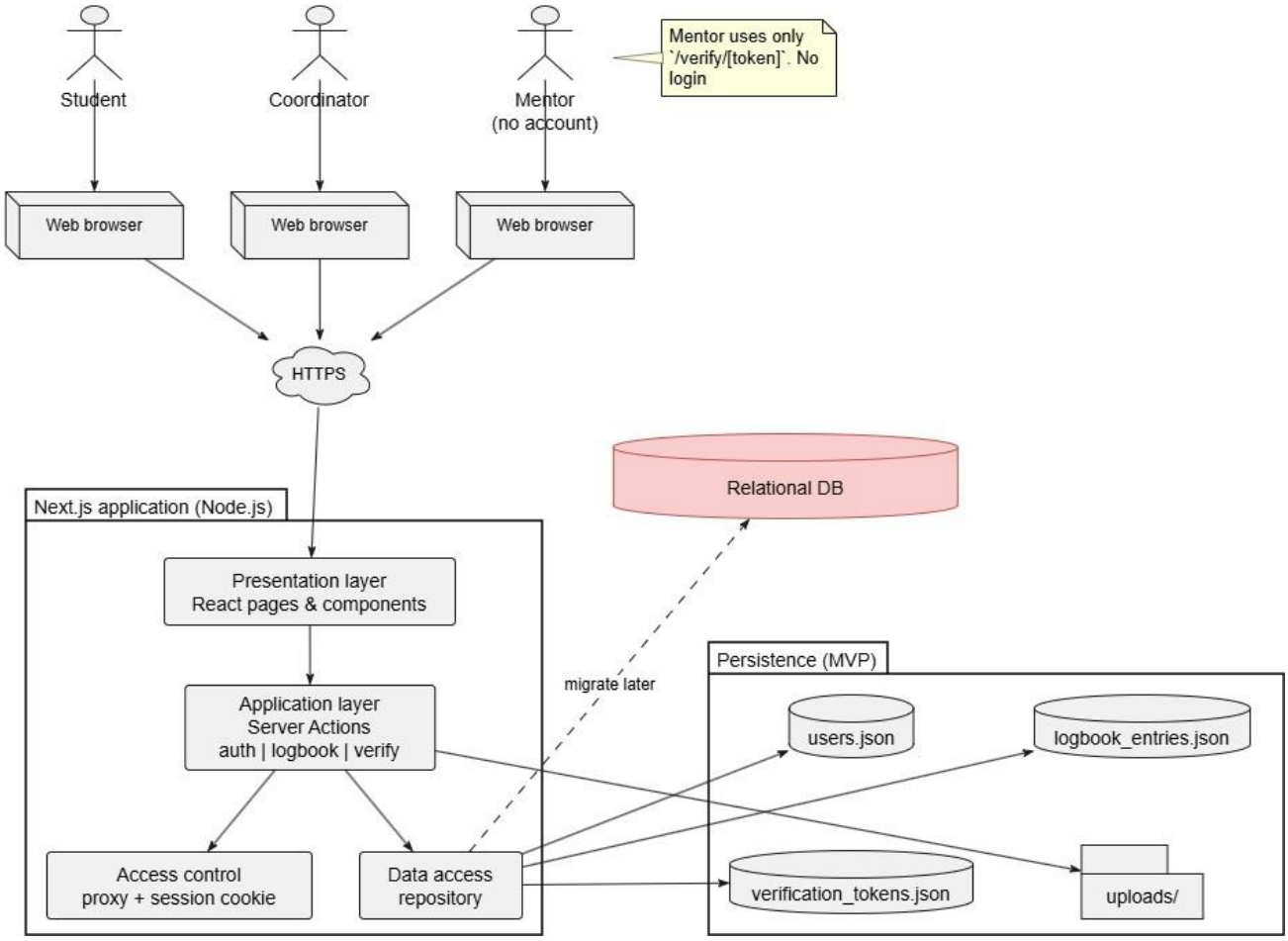


Figure 1 Architecture Diagram

4.3 UML Design

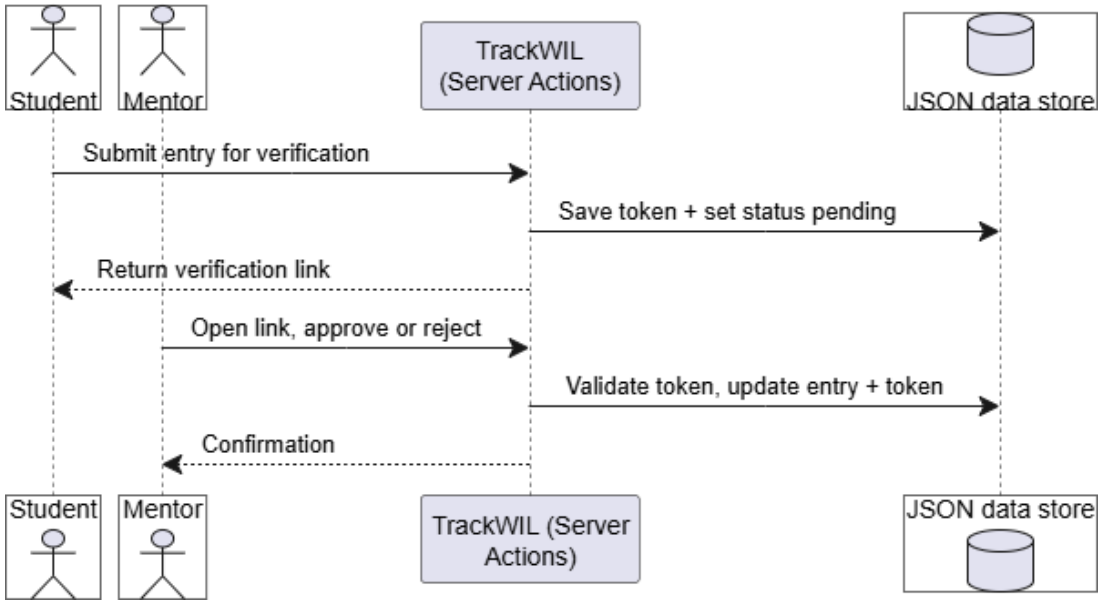


Figure 2 Sequence Diagram

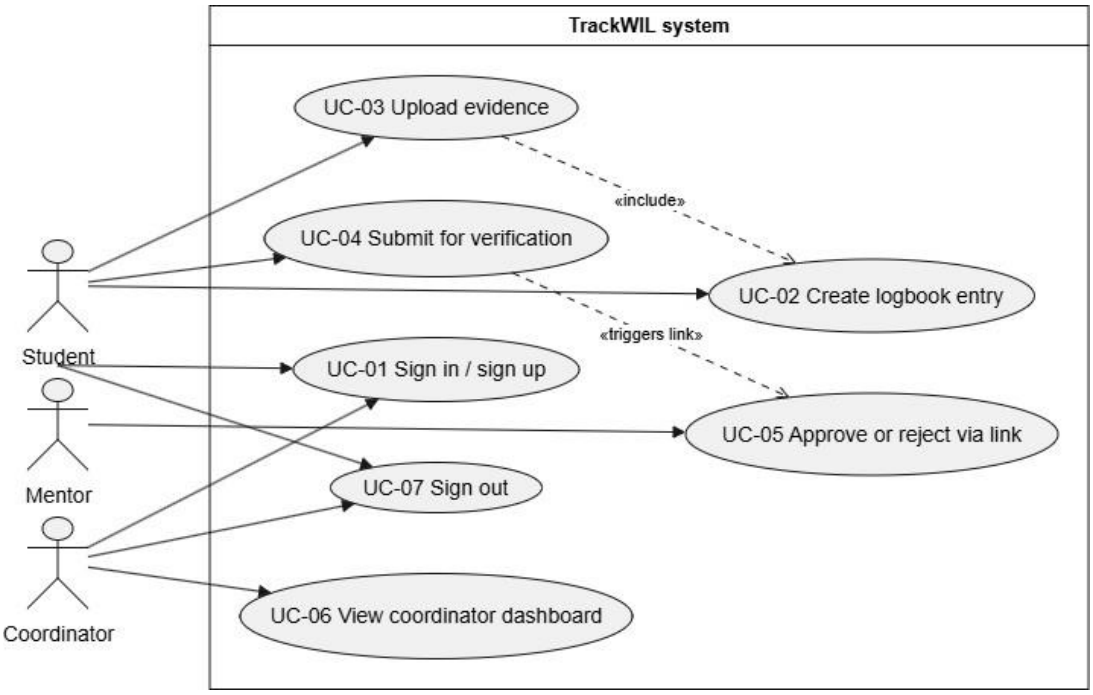


Figure 3 Use Case Diagram

The sequence diagram shows mentor verification as message between the student, the mentor, TrackWIL, and the database.

The use-case diagram has 3 actors:

- Student – signs up, logs hours, uploads evidence, and shares the entry for checking.
- Mentor – Approves/rejects a student's entry.
- Coordinator – Monitors progress across students.

4.4 Database Design

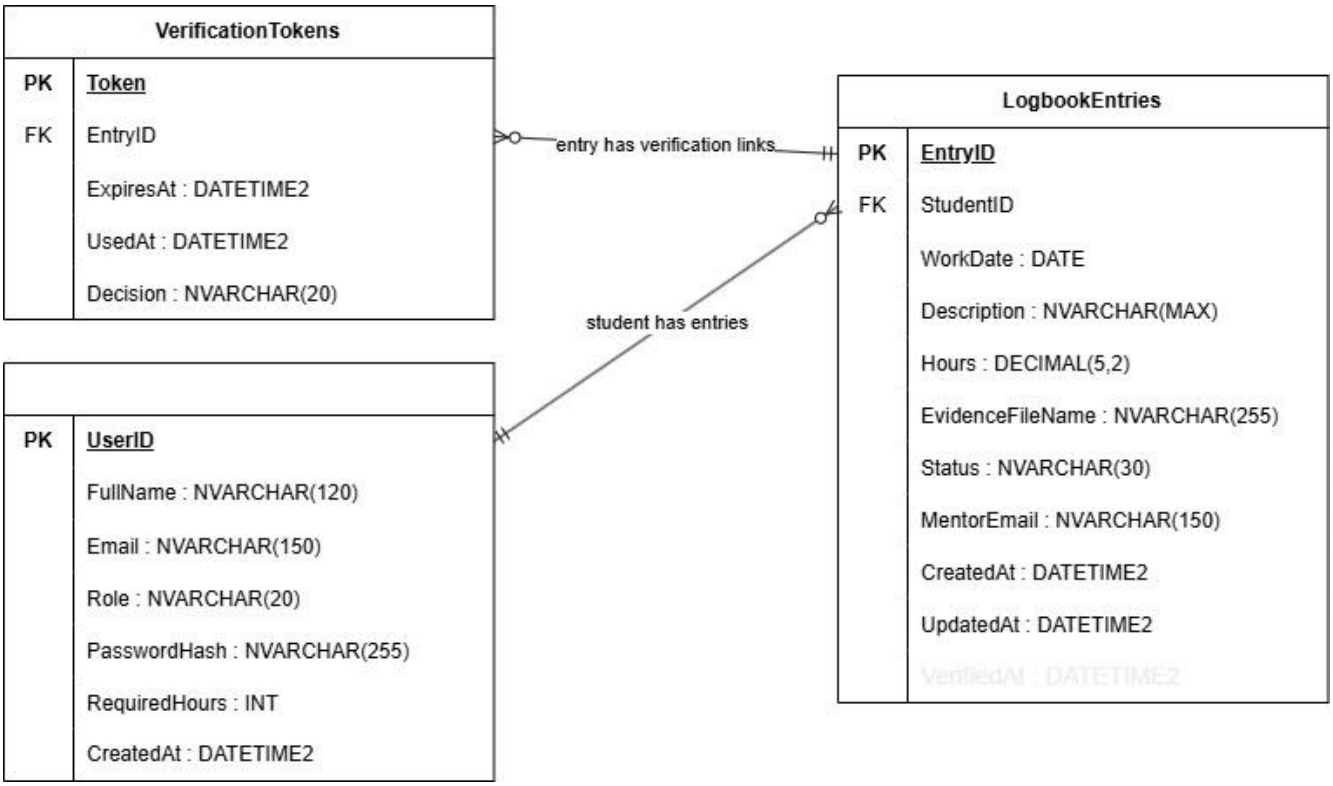


Figure 4 ERD

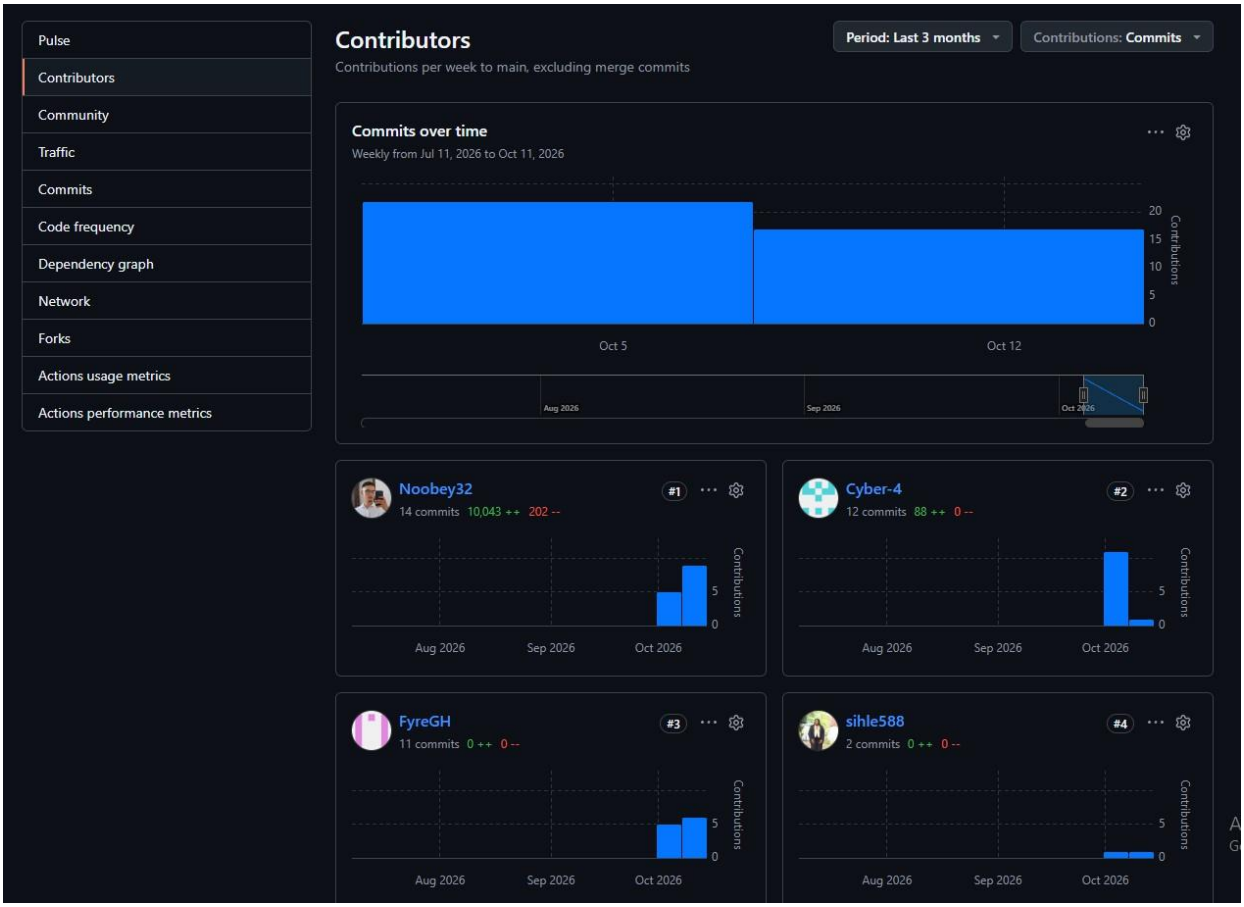
4.5 User Interface Design

Show wireframes or prototypes and explain usability decisions.

4.6 Development Environment

- Languages: TypeScript, HTML and CSS through JSX, SQL in the schema script
- Framework: Next.js 16
- IDE: Visual Studio Code
- Database: JSON file system
- Libraries: Tailwind CSS
- Cloud services: none. Local only
- Version control: Git and GitHub (https://github.com/Noobey32/TrackWIL)

Development Evidence



Revised Chapters 1 and 2

Feedback Received	What your team changed	Where
Remove template formats	Removed template front covers	Chapter 1, P1 Chapter 2, P1
Remove template formats	Removed heading descriptions	Under each heading
Fix scope inconsistencies	Removed digital signature from scope	Chapter 1, P3-4
Fix scope inconsistencies	Corrected status message inconsistency	Chapter 2, P4

WEEKLY PROJECT PROGRESS REPORT

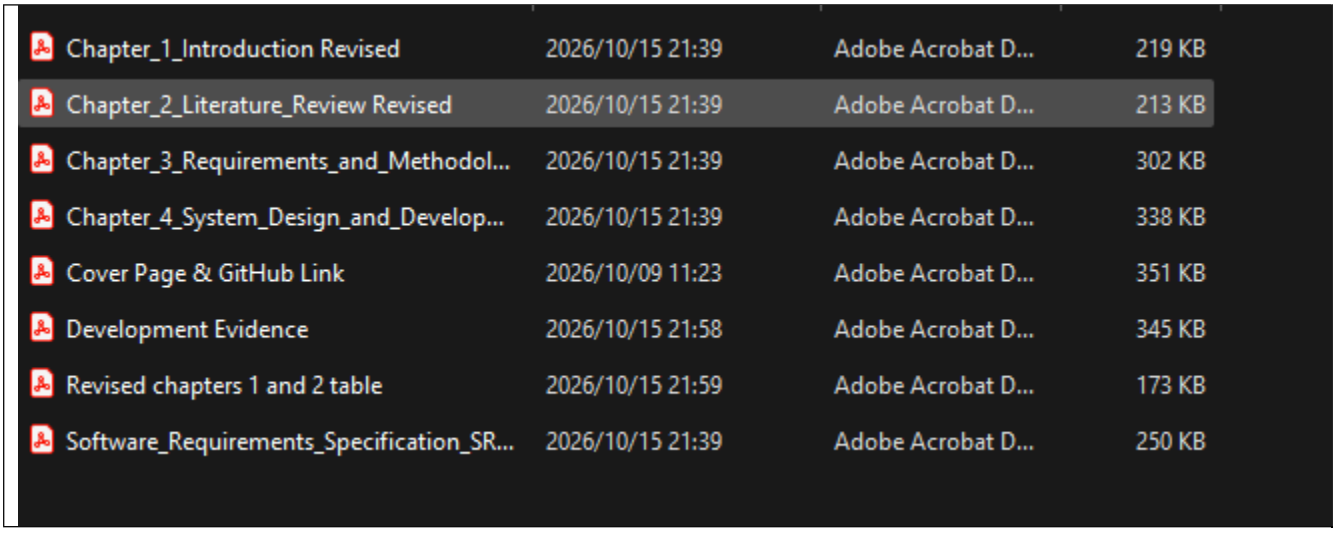
Week / Date

Week of the 12th – the 16th of October 2026

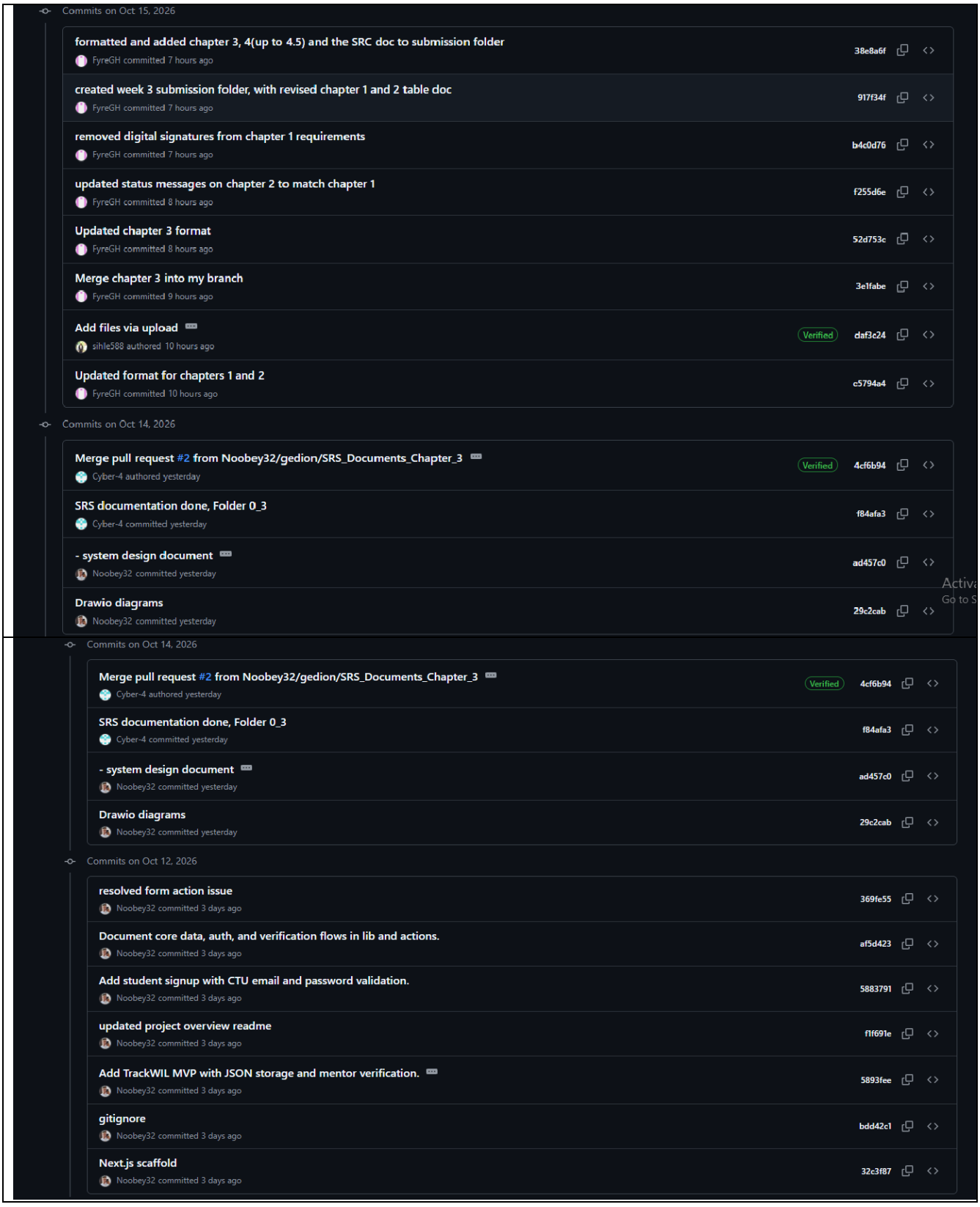
Planned Work

- Chapter 3: Requirements and Methodology
- Software Requirements Specification (SRS)
- Chapter 4: System Design (4.1 to 4.6)
- Development Evidence
- Revised Chapters 1 and 2
- Weekly Progress Report
- Included Work plan
- Actual Development of the Application

Completed Work



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Commits on Oct 15, 2026
formatted and added chapter 3, 4(up to 4.5) and the SRC doc to submission folder
FyreGH committed 7 hours ago
38e8a6f
created week 3 submission folder, with revised chapter 1 and 2 table doc
FyreGH committed 7 hours ago
917f34f
removed digital signatures from chapter 1 requirements
FyreGH committed 7 hours ago
b4c0d76
updated status messages on chapter 2 to match chapter 1
FyreGH committed 8 hours ago
f255d6e
Updated chapter 3 format
FyreGH committed 8 hours ago
52d753c
Merge chapter 3 into my branch
FyreGH committed 9 hours ago
3e1fabe
Add files via upload
sihle588 authored 10 hours ago
Verified
daf3c24
Updated format for chapters 1 and 2
FyreGH committed 10 hours ago
c5794a4
Commits on Oct 14, 2026
Merge pull request #2 from Noobey32/gedion/SRS_Documents_Chapter_3
Cyber-4 authored yesterday
Verified
4cf6b94
SRS documentation done, Folder 0_3
Cyber-4 committed yesterday
f84afa3
- system design document
Noobey32 committed yesterday
ad457c0
Drawio diagrams
Noobey32 committed yesterday
29c2cab
Commits on Oct 14, 2026
Merge pull request #2 from Noobey32/gedion/SRS_Documents_Chapter_3
Cyber-4 authored yesterday
Verified
4cf6b94
SRS documentation done, Folder 0_3
Cyber-4 committed yesterday
f84afa3
- system design document
Noobey32 committed yesterday
ad457c0
Drawio diagrams
Noobey32 committed yesterday
29c2cab
Commits on Oct 12, 2026
resolved form action issue
Noobey32 committed 3 days ago
369fe55
Document core data, auth, and verification flows in lib and actions.
Noobey32 committed 3 days ago
af5d423
Add student signup with CTU email and password validation.
Noobey32 committed 3 days ago
5883791
updated project overview readme
Noobey32 committed 3 days ago
f1f691e
Add TrackWIL MVP with JSON storage and mentor verification.
Noobey32 committed 3 days ago
5893fee
gitignore
Noobey32 committed 3 days ago
bdd42c1
Next.js scaffold
Noobey32 committed 3 days ago
32c3f87

Challenges / Issues

The time constraints are a bit tuff, with all the assessment, the studying, coordinating on everybody's schedules etc.

Next Week Plan

- **Monday -** Striving to better the MVP.
- **Monday -** Chapter 4 completed, including wireframes, and 4.6 - 4.8
- **Tuesday -** Chapter 5, Testing, Results & Evaluation (draft)
- **Tuesday/ Wednesday -** Testing Workbooks
- **Wednesday/ Thursday/ Friday -** Full dissertation draft
- **Thursday/ Friday -** Working on a presentation pan.

Supervisor Feedback

Comments, corrections and go-ahead.

_____

_____

_____

_____

Sign-off

Group Representative: _____ Supervisor: _____ Date: _____

_____

_____

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PROJECT CHARTER & WORK PLAN

Project Overview

TrackWIL replaces lost PDF and email workplace logbooks for occupational students at CTU Cape Town.

Students, host employers, and coordinators need one place to record, verify, and see required hours.

The core product is due on 29 October 2026: logbook entry, token verification, and a coordinator dashboard.

Aim and Objectives

The aim is to build and evaluate TrackWIL so coordinators can see hour compliance without a manual file audit.

Objective 1 is logbook entry with evidence and progress. Objective 2 is employer sign-off by a secure token link. Objective 3 is a dashboard of on-track, awaiting, and overdue logbooks.

Scope and Deliverables

In scope are the three core features, the dissertation chapters, ethical clearance, GitHub evidence, tests, and the demo.

Analytics, bulk exports, audit logs, and staff or student data collected before written campus permission are out of scope.

Deliverables are the web app, this charter, the weekly submissions, the dissertation, and the 29 October demonstration.

Team and Roles

Willem Basson is group lead and owns client contact, the repository, milestones, and ethical clearance.

Sihle Kayise

Dylan Shepard is in charge of QA so that the project reaches its requirements.

Moegammad Green is the developer.

All four members share design, the core build, testing, and the final dissertation and demonstration.

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Milestones

9 October 2026: submit Chapters 1 and 2, this charter, ethical clearance, and the GitHub repository link.

16 October: requirements and design.

23 October: core build and tests of logbook, token, and dashboard.

29 October 2026: dissertation, working app, evidence folders, and demonstration.

Risks and Assumptions

Permission to use staff or student data may arrive late, so evaluation uses synthetic data until POPIA consent exists.

The scope risk is treating analytics, bulk exports, or audit logs as required work before the core product is done.

If a Friday milestone slips, the owner tells Willem Basson that week and the group cuts back to the three core features.

Approval

This version 1 charter is the Week 2 plan submitted for supervisor review on 9 October 2026.

Willem Basson signs for the group as lead; Moegammad Green prepared this work plan.

Ms Amenda Raganya records supervisor approval after the Week 2 review.

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Work plan

Week of 9 October 2026: start and evidence

Task	Owner	Done when
Submit Chapters 1 and 2	Willem Basson	Chapters are in the evidence folders
Submit this charter and work plan	Moegammad Green	Charter is ready for supervisor review
Ethical clearance request	Willem Basson	Request is submitted; no staff or student data is collected before written permission
GitHub repository and link	Moegammad Green	Repository link is recorded for the submission
Confirm core scope with the group	All	Logbook, token, and dashboard are the only required product work

Week of 16 October 2026: requirements and design

Task	Owner	Done when
Use cases for student, host employer, and coordinator	Sihle Kayise	Each core feature has a use case the group agrees on
Activity flow for entry, token sign-off, and dashboard status	Dylan Shepard	The three flows are drawn and match the scope
Data needed for hours, evidence, token, and status	Moegammad Green	Fields are listed without storing real personal information
Screen outline for the three features	All	The group can demo the intended screens on paper or in a sketch

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Week of 23 October 2026: core build and tests

Task	Owner	Done when
Logbook entry with evidence and hours completed	Willem Basson	A student can create and view an entry and see progress
Time-limited token and employer sign-off with no account	Moegammad Green	A host can open a link, verify, and the token expires
Coordinator dashboard: on-track, awaiting, overdue	Sihle Kayise	A coordinator can see those three states from synthetic records
Tests of the three core features	Dylan Shepard	Each feature has a recorded pass or a listed defect
Fix defects that block the demo	Moegammad Green, with Dylan Shepard	The three features run together for the demonstration

Week of 29 October 2026: submit and demonstrate

Task	Owner	Done when
Dissertation chapters and evidence folders	Willem Basson	Folders match the module submission list
Working app on synthetic data	Moegammad Green	Demo path is logbook, token, then dashboard
Test evidence	Dylan Shepard	Test notes sit with the build evidence
Demonstration	All	The group shows the three core features to the supervisor

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